



Paper ID : 261374

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Subject Code: MPH201T

Roll No:

**MPHARM****(SEM. II) THEORY EXAMINATION 2025-26****MOLECULAR PHARMACEUTICS (NANO TECH & TARGETED DDS)****TIME: 3 HRS****M.MARKS: 75****Note:** 1. Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt *all* questions in brief.****10 x 2 = 20**

|    |                                           |
|----|-------------------------------------------|
| a. | Define drug targeting.                    |
| b. | What is active targeting?                 |
| c. | Define liposomal drug delivery system.    |
| d. | What are microcapsules?                   |
| e. | Define aquasomes.                         |
| f. | What are phytosomes?                      |
| g. | Define intranasal drug delivery system.   |
| h. | What are non-viral gene transfer systems? |
| i. | Define antisense molecules.               |
| j. | What is biodistribution?                  |

**SECTION B****2. Attempt any *two* parts of the following:****2 x 10 = 20**

|    |                                                                                                                     |
|----|---------------------------------------------------------------------------------------------------------------------|
| a. | Discuss tumor targeting and brain targeting with suitable examples.                                                 |
| b. | Explain types, preparation methods and evaluation parameters of microcapsules and microspheres.                     |
| c. | Describe pulmonary drug delivery systems including aerosols, propellants, container systems and evaluation methods. |

**SECTION C****3. Attempt any *five* parts of the following:****7 x 5 = 35**

|    |                                                                               |
|----|-------------------------------------------------------------------------------|
| a. | Write a note on targeting methods in drug delivery systems.                   |
| b. | Discuss preparation and evaluation of liposomes.                              |
| c. | Explain preparation and applications of monoclonal antibodies.                |
| d. | Write a short note on niosomes.                                               |
| e. | Discuss evaluation parameters of pulmonary drug delivery systems.             |
| f. | Explain liposomal gene delivery systems.                                      |
| g. | Write a note on therapeutic antisense molecules and aptamers as future drugs. |